A Galaxy Law Tied to Dark Energy, and a Map of What It Leaves Unexplained

Working note (PAPER67), version 1.2, 2026-11-04. AI-assisted research programme; not peer reviewed. $\kappa = \frac{1}{2}$ is fitted. The theory is not closed. Nothing here says the data favour this framework over any other.

Abstract

We report what is left of a research programme after its own final grade. A galaxy law, $g_N = g^2/(g + a_0)$ with $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ and $\kappa = \frac{1}{2}$ fitted, describes isolated disc galaxies well on the canonical footing: SPARC rms 0.0541 dex at median Υ 0.52, no tilt with speed, no slope with surface brightness at fixed speed. The law is a restatement of MOND's simple form with a proposed origin for its scale. Earlier authors proposed that origin, and the scale's tie to dark energy, as distinct from cH_0, has no decisive test yet.

The law fails IC 1613, several Local Group classes, cluster shear and X-COP, whose hydrostatic masses it misses by a factor of 2.93, and the unscreened Solar System. Its Milky Way result depends on a vertical form it doesn't supply, and at 60–150 $\mathrm{km\,s^{-1}}$ it asks for about twice the stellar mass estimated from photometry. Its one action, v3s, is adopted only in a limited scope (below), and the law supplies no dust-like term for the CMB. To cover those we developed a cold dark energy sector. On measured data every graded candidate fails, including the law alone, and the sector is now retired. We publish the scorecard, the two walls that stopped it, and twelve requirements any successor must meet. This is numerical evidence for the families tried, not a no-go theorem. We close with dated tests that can still kill the law.

Version 1.1 adds lanes run after the retirement (Section 7). None was blind, all were run by AI subagents, and no outside group has reproduced them. A gas term fitted on X-COP reaches 14 of 15 keys on the canonical footing, fragile, never 15, and tied to present gas it is dead on the Bullet. A repaired test gives no statistical support that the a_0 formula is special; that doesn't refute it, and the test lacks power. A line of v1.0 about v9's α_1 is corrected. A published rise of a_0 with redshift (MUSE-DARK III), which v1.0 omitted, counts against the law's flat band at face value; we read it as mass-route dependent and not decisive, and we say what that reading rests on. Formula searches found no hit. A definition audit moves single lines of the scorecard, and each candidate still fails at least two on every reading run. Lanes through 2026-11-04 add four results. No unscreened action gives the law and passes Cassini; only the frozen screened theory's screening length passes the Solar System, and that theory still fails cosmology. The record's lead for the Milky Way and Cassini together, a phantom mass smoothed over a length, has no surviving form once the full 3D field is used. The law's lensing shortfall, about 0.09 dex inside 0.3 Mpc, survives every estimator on record and the measured hot gas. A light fermion beside the law, tried under a one-lane lift of the no-particle rule, fails the early universe.

Version 1.2 records the owner's decision of 2026-11-04, which superseded a no-action call made the same day: v3s, a screened Khronon theory, is adopted as the action for the a_0 law, scoped to isolated galaxies, the Solar System, galaxy lensing and gravitational waves. It supplies no cluster or CMB mass, its failures there stand, two of its checks are undecided on numerics, and a_0 is still put in by hand.

1 Scope

This is a programme report, not a complete theory. Four owner decisions on the record set its form: the a_0 floor was removed from the cold sector (2026-10-31), the search for a derived threshold stopped (2026-11-01), the cold sector was retired (2026-11-01), and v3s was adopted as the law's action in a stated scope (2026-11-04; Section 5). Earlier deposits stand as dated records. The two footings, canonical $a_0 = 9.3603 \times 10^{-11}$ and alternative $1.1312 \times 10^{-10}\,\mathrm{m\,s^{-2}}$, are never pooled. This paper's audit script checks the quoted numbers against committed files. It doesn't check the prose.

2 The law, and whose idea it is

$$g_N = \frac{g^2}{g + a_0}, \qquad a_0(z) = \kappa c\sqrt{G\rho_{\rm DE}(z)}, \qquad \kappa = \tfrac{1}{2}\ \text{(fitted)}.$$

Credit. The first relation is Milgrom's modified dynamics in its simple form [1]. The numerical nearness of a_0 to cH_0 and to $c^2\sqrt{\Lambda}$ is his observation [1, 2]. Limbach, Psaltis & Özel [3] proposed the tie to the dark energy density. None of this originated here.

κ isn't derived. Of 550 derivation or selection attempts on the record, none derives $\kappa = \frac{1}{2}$. The record's closure (KC1) reduces every route to one missing length: $\kappa = \frac{1}{2}$ is the same as a length of exactly $c/\sqrt{G\rho}$, and no field

equation on record supplies it. A screen of 16 further premises found none that does (KC2). It isn't singled out by SPARC either. A profile likelihood over 175 galaxies has a free best fit of 1.0766×10^{-10}. Against it, $\kappa = \frac{1}{2}$ on $\rho_{\rm DE}$ reads $\Delta\chi^2$ 63.90 (1.82σ with galaxy-clustered errors), the alternative footing 7.04 (0.60σ), and Milgrom's own $cH_0/2\pi$ 5.34 (0.53σ). So Milgrom's value fits at least as well as ours. Data can't settle the coefficient yet. Telling $\frac{1}{2}$ from 0.461, the value a 2π horizon form gives, needs a_0 to 2.8 % at 3σ. The shared galaxy-mass floor is 9.47 %, and the best case with $\rho_{\rm DE}$ is about 5 % (KC1). Only new observations, not more archive data, would lower it.

What this programme adds: a reconstruction of $a_0(z)$ from raw BAO and supernovae with a stated ruler, the graded comparison below, and the failure record.

The band, and what it rests on. The reconstruction gives $a_0(z)/a_0(0)$ of 0.91 [0.76, 1.03] at $z = 1$, 0.94 [0.71, 1.12] at $z = 1.5$ and 0.91 [0.33, 1.21] at $z = 2.33$ (68 %), against 1.76, 2.35 and 3.49 for the rival $a_0 \propto H(z)$. These ratios are the law's predictions from the measured $\rho_{\rm DE}(z)$, not measured galaxy values. Constant dark energy isn't rejected (1.97σ, primary). **This band isn't free of the retired sector's input:** it takes $\rho_m(z) = \omega_m(1+z)^3$ with a cold mass from the CMB, and with positivity alone, no CMB and no H_0, its check fails. The law doesn't explain that matter term (Section 5).

The distinctive claim has no decisive test. At $z \approx 0$ the law can't be told from $a_0 \propto cH_0$. Separating them at 3σ needs a_0 to 0.055 dex at $z = 1$ and 0.084 dex at $z = 2.33$. The best public sample reaches 0.4–0.5 dex. A frozen decision rule (AT1, its arithmetic checked in Lean) puts the binding term in a sample's common mode, the part of its mass error that doesn't cancel against a local reference. That must sit under the same 0.055–0.084 dex. At 0.05 dex, with 0.14 dex per galaxy, 41 galaxies decide at $z = 1$ and 5 at $z = 2.33$. At 0.075 dex only $z = 2.33$ stays open, and even a sample that passes the rule's power gate can land inconclusive. Applied to every high-z sample on disk, 22 samples in 52 cells, the rule leaves all 52 BLOCKED, with common modes of 0.30–0.60 dex and no same-estimator local reference (AT2).

A published rise that counts against the flat band at face value. MUSE-DARK III [13] fits the radial acceleration relation in 79 galaxies at $0.33 < z < 1.44$ and reports $a_0(z \sim 1) = 2.38^{+0.12}_{-0.10} \times 10^{-10}\,\mathrm{m\,s^{-2}}$ and, for $a_0(z) = a_0(0) + a_1 z$, $a_1 = 1.59 \pm 0.10$ in the same units (95 % intervals). Our band at $z = 1$ on the canonical footing is $0.91 \times 9.3603 \times 10^{-11}$, a factor of about 2.8 lower. Alcock [14], an independent author, fits the survey's four bins as read from its figure: a constant a_0 is excluded at 5.3σ if the bars are 1σ and at $\geq 10\sigma$ if they are 95 % intervals, the reading the survey's stated convention for its main fits points to, statistical errors only. On that reading $a_0 \propto E(z)$ is itself in tension (4.2σ). The Letter notes that frameworks tying a_0 to the cosmological constant are under the same pressure. That is this law. It was received 2026-07-10, before our first deposit using these data (2026-09-06), and we claim no priority on it.

Why we read it as non-diagnostic, not as a pass. The survey's main fit leaves stellar mass free of its SED estimate. Our own frozen run (CFG262, a different sample of 109 galaxies, not a direct a_0 measurement) finds the rise travels with the mass route: the implied change from z 0.52 to 1.20 is $+0.570 \pm 0.094$ dex on fitted masses, -0.115 ± 0.489 on SED plus H_2, and $+0.036 \pm 0.138$ on SED alone, with a spread of $\times 12.7$ in level in the top third. The SED routes are consistent with flat and also with $H(z)$, so they rescue nothing. A redshift-dependent mass drift of 0.13–0.19 dex across the bins would remove the rise; the survey puts its unmodelled molecular gas near 0.2 dex, but nothing here shows that drift is present. MUSE-DARK II [15] finds no baryonic Tully–Fisher zero-point evolution at $z \sim 1$ (0.00 ± 0.06 dex), with a further ± 0.16 dex from the local zero point in its own model comparison. On the survey's masses, the band and the dark energy tie fail. That reading stands until a per-galaxy reanalysis with independent baryon masses says otherwise.

3 What it does in isolated galaxies

A caution on the Υ line. The 0.50–0.80 line was first written for this law, and its lower edge is a population model's central value, not a measurement. So passing it is weaker than it looks; Section 4 sets it against measured masses.

Surface brightness is the strongest result. At fixed speed, a halo set by mass alone can't follow the disc: every version of the retired sector's halo tilts, -0.17 to -0.29 per dex, and the law doesn't. That's a real discriminator on the data used, between the law and these halos only.

Isolated-galaxy lensing is shape, not a clean amplitude. KiDS-1000 prefers the law's shape over the retired halos ($\Delta\chi^2$ +366.0 against the best native halo). But the law's amplitude depends on colour and on the baryon

Table 1: The law on isolated systems. FG1 is the frozen final grade; a clean-room regrade (CRG1) reproduced its table, 289 match, 6 close, 0 disagree, mainly a test of code.

line	canonical	alternative	lane
SPARC rms, median Υ (line 0.50–0.80)	0.0541, 0.52	0.0537, 0.44 **fails the line**	FG1
tilt with speed	+0.02	−0.02	FG1
LITTLE THINGS, median E	−0.025	−0.064	FG1
63 groups, Z	+0.31	−0.31	FG1
slope with surface brightness, 60–120 km s^{-1}	-0.016 ± 0.061, flat	-0.023 ± 0.063, flat	HS1
local a_0, MeerKAT against SPARC	+0.032 dex (line 0.15)		PAPER60

inventory: blue lenses read $A = 1.016 \pm 0.086$ in one lane and 0.878 in a later forward model that adds a hot gas inventory, and red lenses 1.587 and 1.313. No tested model fits these profiles in absolute terms. Inside 0.3 Mpc the data sit above the law under every grader on record, by +0.0887 dex (alt +0.0500), and no single swap of estimator brings the full-profile grade within $|Z|$ 2 (LE1, ESTIMATORS DISAGREE, NO DEFECT). The hot gas measured through kSZ doesn't close it (CW6, GAP UNCHANGED). That gas sits mostly at 200–500 kpc. Red lenses move from 1.313 to 1.306, still 4.87σ above 1, blue from 0.878 to 0.861, and the full-profile Z worsens from +9.27 to +10.45 (alt +6.52 to +13.97). Gas held inside the virial radius narrows red, but one gas rule for both colours can't bring red within 2σ without pushing blue over.

4 Where the law fails

- **Milky Way.** The law gives only the in-plane force, so the vertical test needs a form it doesn't supply. The field-equation form (AQUAL) is INFEASIBLE, 0 cells, on both footings. So is a real QUMOND solve, which differs from AQUAL by at most 0.13 % in $K(R_0)$, and varying heights and thick-disc share doesn't reach the line: in priors the lowest is 75.9 (alt 76.2) against its top of 70.8 (UX5, K_z FAILURE SURVIVES). That rests on a kinematic summary fitted inside Newtonian potential families. The round rule, which has no field equation behind it, passes in 1 in-prior cell (alt 6) (LW1), 72 (alt 196) with a shorter thick disc, and there it still misses the K_z scale length. On an older, weaker gate, 7 of 175 cells pass (alt 4), all with a short disc and the lightest local column; those have $K_z(R_0)$ 76.7–80.9 against 66.4 ± 2.2, +4.7 to $+6.6\sigma$, and leave the outer slope excluded (Z −3.88). FG1 counts this as FAIL. A filter line tried to pass the Milky Way and Cassini together by smoothing the law's phantom mass over a length (KT14–KT32, CZ1–CZ5). No form survives. Fixed widths fail Cassini or the planets (CZ2, CZ3). Tidal rules fail the Milky Way once the vertical tide is in (KT29–KT32). The local rule fails Cassini on the 3D field, 13.3 times the ceiling (alt 18.1), because the Sun's and the Galaxy's pulls cancel at a saddle where its smoothing length goes to zero (CZ5). Its earlier passes rested on a 1D approximation. The first Milky Way pass also needed a thick disc heavier than star counts give: at their share of 0.12, 0 of 450 cells pass (KT18).
- **IC 1613.** $E = -0.593$, $Z = -7.9$; it fails under every halo-free inclination. One galaxy: disfavoured, not excluded. $Z = -7.9$ has no velocity error; with 15 % on V the base is −3.94, and −2.35 and −2.83 at RF2's central inclinations 34.4 and 37.9. Only inclination could close it: $E = 0$ needs i 22.1 (alt 21.1). A disturbed gas disc is a live explanation, not computed (UX4).
- **Local Group classes.** Z: MW ultra-faints +5.22, bright M31 +2.20, other M31 +2.29, MW classicals +1.69, field +0.57. Adding the external field makes it worse: ultra-faints +6.76, bright M31 +3.97. And at fixed M_V the data are flat in host field, -0.05 ± 0.19 dex per dex, where AQUAL with its external field predicts −0.76, so neither version describes these satellites (LW1). Listed systematics don't close them (UX4, FAILURES STAND, both footings). A central stack of binaries, tides, membership and heating leaves ultra-faints at Z +5.08 and other M31 at +2.75; bright M31 reaches +1.47 only with an assumed orbit nobody measured. A ceiling stack overshoots and breaks what passes, field Z −7.67. The field's pass doesn't test this regime: 41 satellites are fainter than the faintest field dwarf with a dispersion. The filter rules don't lift the dwarfs: on the 3D field the local rule passes 17 of 24 Local Group dwarfs, as plain QUMOND does on canonical (alt 18) (CZ5). A sweep of every external field inside ν on record found no defect in these rows or in the Solar System

line below (CW3).

- **Measured stellar masses, 60–150 km s^{-1}.** The law's Υ is 0.62 and 0.65 where GSWLC stellar masses give 0.277 and 0.304 (Z +3.4, +5.6). These are estimators against W1 luminosity, not a 3.6 μm measurement. Gas dominance doesn't excuse the 100–150 bin. Offsets, distances and inclinations don't carry it (UX5, NOT CARRIED). The level miss against GSWLC is narrow, best $|Z|$ 2.12 at +0.204 dex; Taylor needs more than a Salpeter shift; and no offset changes the step between 100–150 and 150–200, which rests on 23 and 25 galaxies. Galaxies with accurate distances still read 0.622±0.063.
- **Disc weighings.** On this law's own form, DiskMass σ_z reads round rule $r = 1.239$ (alt 1.204), outside 0.90–1.10, a FAIL as frozen (LW1). Earlier lanes read $r = 1.250$ and 1.342 with a different interpolating function. Same-population dispersions [7, 8, 9] correct r by c of 1.414–1.517, giving 0.824–0.884 one-zone and, with exact Jeans and measured thickness, inside the line. But the sources' own tables run c from about 1.24 at the relevant radius to 1.88 in integrated light, so the question is open (OD1, NOT SEPARATED). NGC 628 reads $\Upsilon_{3.6} = 0.75 \pm 0.18$ and NGC 6946 about 0.4.
- **The Solar System.** The unscreened law fails Cassini's Q_2 by 2.44–4.59 times the 2σ line (alt 2.85–5.19). The pass in earlier reports belongs to a screened covariant theory. That line's final verdict (v3s) was that it holds as a galaxy law and fails as gravity plus cosmology. So the law as written here needs a screen it doesn't contain. A high-precision rerun cleared the one failed check behind the screened pass, so that theory's Solar-System gate is clean on both footings, Q_2 at least 4.8×10^7 (alt 5.7×10^7) inside the line (C6P). Its verdict stands. The owner has since adopted v3s as the law's action in a limited scope, so within that scope the screen is v3s's (Section 5).

5 What it leaves unexplained

- **Cluster cores and shear.** From measured gas, in 318 eROSITA clusters with raw KiDS shear, the law is short by $A = 1.91 \pm 0.12$ at 0.5–1.2 Mpc (alt 1.75 ± 0.11) and matches at 1.2–3 Mpc, $A = 1.01 \pm 0.09$. On the measured chain it fails 10 of 15 keys on both footings, G-OUT-M3 +4.76, joint p 3e-8. On this statistic that is worse than the retired native profile (LW1).
- **X-COP.** The law's own number is +0.467, rms 0.476: $M_{\rm HSE}/M_{\rm law}$ of 2.93 (alt 2.72), all 7 clusters short. Taking 0.12 dex off the hydrostatic masses still leaves 2.23 (alt 2.06). An earlier scorecard showed another rule's row here; LW1 corrects it.
- **What is missing in clusters, as a description.** Inside about R_{500} the deficit is near 8 times the gas at 0.2–0.5 R_{500} and 5 times at 0.5–1 R_{500}, with no detected mass dependence, no BCG link and no preferred scale found, picked from 31 looks (LM1). Trained on KiDS, it predicts X-COP's hydrostatic masses out of sample, but HSC lensing reads it about 1.7 times higher, and tracking gas can't be told from tracking M_{500}, so its frozen label is NOT (GM1, which issued labels after a failed check and says so). No mechanism follows.
- **Outer shear against gas mass.** $\Delta\Sigma_{\rm OUTER} \propto M_{\rm gas}^s$ with $s = 0.90 \pm 0.21$, against the law's 0.47 and Newton's 0.84. The amplitude agreement above is an average over a slope the law misses.
- **The CMB and growth.** a_0 is absent from the linear FRW equations, so the law can't act on the background. It can act on the perturbations. The potential that CMB lensing maps at $z = 0.5$–2.33 has gradients of 0.003–0.22 a_0 (Planck PR4; WG1), inside the law's range. Run from a baryons-only start with no cold term (WG2), the law gets CMB lensing wrong in shape, not only in size. Off on the web, with the measured expansion, it gives under a thousandth of the measured lensing at every multipole. As QUMOND on the peculiar field, it gives about 14 times the measured value at $L \approx 28$ but 0.08 at $L \approx 377$, converged from 128^3 to 192^3 (WG6's shape check). A baryons-only start is Silk-damped, so it has little small-scale potential, and the law boosts the largest scales most. No rule tried fixes both ends. The record's one web rule, a switch that turns the law on only past turnaround, never turns on from this start (WG3). Letting a_0 track DESI's evolving dark energy, as the law says, lowers the whole spectrum by about half without changing its shape (WG4). A screening term put in by hand on the law's extra mass removes the large-scale excess but leaves the spectrum 1.5 to 7 times low at $L \gtrsim 100$ (WG6). An earlier reading, that halving the law's extra mass would land, came from grading one amplitude and is withdrawn. WG2's labels rest on one change, made after the results, to an initial-conditions check. The record has no CMB pass without a dust-like component. With one, a window exists at $z_{\rm on} \in [3 \times 10^4, 10^6]$ on the channels run. Retiring the sector leaves this unanswered. **The law offers no replacement, and we don't claim it needs none.**
- **The matter term.** The law takes $\rho_{\rm DE}$ from an expansion history whose matter term is several times the baryons. That term is unexplained here.

- **An action, adopted in a scope.** On the owner's decision of 2026-11-04, which superseded the same day's call that no action is selected, v3s is adopted as the action for the a_0 law, *scoped* to isolated galaxies, the Solar System, galaxy lensing and gravitational waves. The gates it rests on are in Table 2. Outside that scope nothing changes. v3s supplies no cluster or CMB mass, and the 10-24 verdict's failures there stand unchanged: cluster cores, linear growth and the CMB, the radiation era, hosted dwarfs and the Milky Way joint fit stay open. The weak spots carried with the adoption are these. V3X's W2 and W4c(b) are UNDECIDED on numerics: W2's ladder reads Richardson $-0.70\,\%$ against the $2\,\%$ line, but its own check K1 misses at an ungraded radius, and W4c(b) finds every $\omega^2 > 0$ while its spectral ladder misses by up to $14.6\,\%$ against $5\,\%$ and its tensor roots sit 1.1e-10 below c against a 1e-10 line. The constraints act instantaneously (C5's cone criterion fails; carried, not graded). Four items are asserted, not shown: post-Newtonian corrections don't flip signs, the Sun is at rest in the khronon frame, the solar interior is uniform, and the theory is well-posed nonlinearly. The lensing gate has little power: on KiDS alone the MUTATE twin is caught at only 1.9σ, and GH1 is what catches it. And $a_0 = \kappa c\sqrt{G\rho_{\rm DE}}$ is put in by hand, not derived from the action, with $\kappa = \frac{1}{2}$ fitted. A smoothed QUMOND with one Gaussian length passes no length on the static gates (FN1b). A length scaled as $L_{\rm MW}(M/6\times10^{10}\,M_\odot)^p$ opens a WINDOW at p 0.30–0.45 (FN6), canonical 3 of 48 grid points, all resting on one Milky Way cell, alt 9 of 48, with no mass shuffle surviving. It is a two-number description, not a derivation, and is not adopted; its $p = -0.5$ row, a Gaussian stand-in for v3s's scaling and not v3s's own equations, fails the dwarf gate, as every $p \le 0.1$ does. Ten fresh selector ideas for a dust part inert in galaxies found no action (FA1): I1, I2 and I6 were run and killed, four were already dead in the record, and I5 and I7 are dead on paper. Attempts on record are this programme's own versions on AeST [4] and Khronon [5], and don't test those theories as published. **Correction to v1.0:** v1.0 said v9, an AeST embedding, fails the preferred-frame bounds with $\alpha_1 = -2(K_B + 2)$. That is a static-branch ratio. A source moving through the aether has $\alpha_1 = \alpha_2 = 0$ at $G = \tilde{G}$ (AS3). v9 stays non-viable, because it then loses the law for moving systems. v3, Khronon with an added term, fails linear growth: the published cap $\mu_K \le 14.8\,h/\mathrm{Mpc}$ sits 24–172 times below what σ_8 needs. v4, AeST with that term, is ill-posed where the aether twists and runs away on FRW for the dust shape tried. v3s, v3 with switched stiffness, is frozen as a galaxy law that fails cluster cores, linear growth, hosted dwarfs and the Milky Way disc. AX1 grades every action class on record against this law. No unscreened class gives $g^2/(g + a_0)$ and passes the Solar System. Those whose static limit is AQUAL or QUMOND miss Cassini's Q_2 by 1.69–5.18 times its 2σ bound. Those that keep the interpolation relativistically fail on gravitational waves, BIMOND's on a source read only in part. Only v3s's screening length passes the Solar System on both footings (C6P). Smoothed and curvature-dependent QUMOND energies with every adjoint term kept (KT33–KT42) find no viable response with the fixed baryons. The one that passes the four Milky Way cuts and Q_2 accelerates gas away from the plane in the outer disc, gives dwarfs in an external field a negative binding virial (1,805 of 2,952 configurations), and keeps nearly a_0 at Mars and Saturn (all 16 cases fail). v3s passes those three checks on both footings (VH1): no outward cell in any Milky Way model, a positive virial in all 1,476 dwarf configurations per footing, and Mars at $Z = 0.052$ (alt 0.047) on the planetary lines. That fixes none of v3s's failures above. VH1 itself adopts nothing, and a_0 enters by hand in every class.

Table 2: The gates v3s's scoped adoption rests on (owner, 2026-11-04), each lane's README the record. Clusters, growth and the CMB are outside the scope and still fail.

gate	result	lane
every action class graded	only v3s's screening length $L_\star$ passes; unscreened classes fail Cassini's Q_2 by 1.69–5.18×	AX1
Solar System	clean PASS; Q_2 margin to Hees $\ge 4.8\times10^7$ (alt 5.7×10^7); Mars 0.173 of its bound	C6P
PPN γ, β	derived from the action, $\gamma = \beta = 1$; remainders $\le$ 9.8e-14 and 1.1e-12 (canonical)	V3W W1
galaxy lensing = dynamics	KiDS isolated $A = 1.129 \pm 0.206$ (alt 1.035 ± 0.187); little power on amplitude	V3W W3
GW speed, galaxy health	$c_T = c$, $\lvert\Delta v/c\rvert \le$ 5.1e-32 (alt 6.2e-32); no ghost or gradient instability	V3W W4a/b
Solar-System health	(a) quadratic action and (c) 3D hyperbolicity PASS; (b) no instability found, UNDECIDED	V3X

6 The cold sector: tried and retired

What it was. A gravitating component with, at linear order, $w = 0$, $c_s = 0$, geodesic motion and an abundance set by an integration constant. A referee will read those as the linear behaviour of cold dark matter, and that's correct; it was never proposed as a particle, and the programme's no-particle rule is unchanged. Its candidates: C1, the native collapse rule N; C2, N with the measured gravitational response; C3, a basin split at $T \approx 10^{12.3}\,M_\odot$; C5, the native shape with the measured response; with C4, the law alone, as the control.

Table 3: FG1, canonical, with C4's X and DM from LW1 and C5 from HM1. P pass, F fail, – not graded, which isn't a pass. No winner is declared, because the lines aren't independent. Other readings of a definition move single cells (DA1, Section 7); the table stays as frozen.

	S	Tl	LT	G	K	X	Ln	Dw	MW	DM	IC	alt
C1 native N	P	F	F	P	F	P	F	F	F	F	F	no a_0, same
C2 N + response	F	F	F	P	F	F	F	–	F	P	–	no a_0, same
C3 basin split	P	P	P	P	F	P	P	P	F	–	F	also S, Dw fail
C4 law alone	P	P	P	P	F	F	P	F	F	F	F	also S fails
C5 native + response	F	F	F	F	F	F	F	F	P	P	F	not listed

S SPARC, Tl tilt, LT LITTLE THINGS, G groups, K 15 cluster keys, X X-COP, Ln KiDS isolated, Dw Local Group, MW Milky Way, DM DiskMass, IC IC 1613. C4's MW cell depends on the vertical form. C5's row is a reading of its numbers: HM1 printed NO VERDICT as frozen because one mass check failed, and an amendment moves no key by over 0.001. FG1's tag [GAS-CONDITIONAL]: at X-COP's gas scale +0.111 the native rule reaches 15 of 15 keys, rescuing K only; the measured scale is $+0.005 \pm 0.017$.

Wall 1: spirals at 150–250 km s^{-1}. Whatever is dust in the CMB falls into galaxies, and a live disc measures it contracting: $f_Q = 0.872$ [0.54, 1.07] (LC1), 0.946 [0.78, 1.08] with the disc grown from turnaround (PZ1). The SPARC side needs net f of about 0–0.2 at the 0.50 Υ line (RM2, six of 75 cells, three fitted numbers). This is the known Tully–Fisher zero-point problem [6] in measured numbers. A cored halo's response is somewhat weaker, $f_B = 0.753$ [0.729, 0.777] (DR8), not enough. Whether the 0.50 line is itself too high is open (Section 4, UL1, OD1); moving it wouldn't clear tilt or LITTLE THINGS.

Wall 2: outer cluster shear on measured gas. G-OUT-M3 reads 1.56 against 1.04, $Z = +4.41$ statistical, about +3.03 to +3.51 with systematics (M3A1). It's not an artefact on anything measurable from disk, nor the model's edge, and native two-halo supplies 3–11 % of +21 to +27 %.

No derived switch. The basin split's T was located, never derived. Three frozen attempts failed: 13.94 for the turned-around mass and 13.54–13.89 for a clock, against a window of 12.19–12.39 in $\log M$, and a candidate action had no link to mass. In shift-symmetric $P(X)$, a turn-on has $c_s^2 < 0$ at all 14 grid points, a result within that family (BOLTZ1).

Table 4: Doors tried in the closing push, by each lane's own label. Full list: the closing ledger.

idea	lane	result
density term in the law	DR1	SPARC BREAKS, CLUSTERS NOT REACHED
phase-space cap on response	DR7	BITES, FAILS [NOISE-LIMITED]
retention replaces the mass line	RET1	DERIVED LINE FAILS, FITTED ϵ FAILS
barotropic halo	HB1	MOVES, FAILS
fresh hunt, 19 routes	MH1	no mechanism survives whole
starting point	SP3	no response-on start reaches Υ's line in the owner's grid
symbolic search	SL1	TRAIN ONLY, 0 survivors
exact actions per object	AR1	MOVES AND FAILS

Gas retention: the mass method decides it, and it isn't settled. A gas-fraction line from hydrostatic masses, fitted without X-COP or shear, takes the native profile from 12 to 14 of 15 keys with X-COP passing, G-OUT-M3 +3.48, +3.09, with frozen label NOT RETENTION by one cell (FR1). Okabe's HSC masses on eRASS1 gas read 0.216 dex below that line, -4.4σ; through the same grader that overshoots, 4 of 15 keys, G-OUT-M3 −2.54, X-COP failing. A second HSC reading, Umetsu on XXL gas, sits on the hydrostatic line instead

(FR3). Neither line passes all 15 keys and X-COP. Both lanes were frozen after the retirement, weren't blind and don't touch galaxies.

Status. Retired by the owner on 2026-11-01. One lane frozen before that decision, a per-galaxy action response (AR1), landed after v1.0's deposit. Exact isotropic actions at each object's own halo and baryons give Υ 0.45/0.46 against Blumenthal's 0.36/0.37, on two backgrounds, so the line stays missed. The lift goes to the wrong bins: tilt +0.22/+0.23, LITTLE THINGS +0.129/+0.122. Keys read 14 of 15 on gas +0.005, X-COP −0.103, rms 0.150, groups +1.50. There is no mass trend in the response. The solver follows live runs at 10^{10} and $10^{13}\,M_\odot$, where Blumenthal over-predicts. The frozen label, MOVES AND FAILS, is conditional: check K1c missed its line, 2.8e-5 against 1e-6, not traced, and K2 and K7 first failed and were fixed by finer grids. No later result can reinstate the sector; only the owner's decision can. What survives from this work as fact, not as a theory: native collapse passes X-COP on hydrostatic masses, +0.013, rms 0.075, and one adiabatic response has the opposite sign in discs and cluster cores.

7 Lanes after the retirement

Read these knowing how they were made. Every lane in this section was frozen after the retirement, between 2026-11-02 and 2026-11-04. None was blind: each read the record first, and several wrote down what they expected. All were run by AI subagents, and no outside group has reproduced any of them. They reinstate nothing. The one replication below (CU1) came from the same chain and mainly tests code.

A gas term on clusters (UX3, CU1). The rule $g = \text{law} \times (1 + \theta\, k\, g_{\rm gas}/a_0)$, with k 13.29 (alt 14.25) fitted on 4 X-COP clusters, takes canonical keys from 10 to 14 of 15, G-OUT-M3 +0.58, +0.35, X-COP −0.016, rms 0.089. G-OUT-all still fails, −2.71, −2.77. Alt reads 10 of 15. At $k = \pi^2$, keys read 12 of 15 and X-COP fails. Unswitched, it breaks discs. A switch that shuts it off in discs passes largely by construction, and can't be told from a "clusters are big" control. There is no action or field equation, and the frozen label is KEYS FAIL. CU1 rebuilt it (73 match, 4 close, 1 disagree of 78) and stressed it. The 14 is fragile: it holds only for k 12.87 to 15.09, and k fitted on all 7 clusters reads 13. Joint p falls from 0.38 to 0.18 at the worse h (alt 0.10). G-OUT-all needs $k \leq 7.2$, G-OUT-M3 $k \geq 8.8$, G-MID-T3 $k \geq 12.9$, so no k gives 15 of 15, and no bootstrap draw does. If hydrostatic masses are 0.12 dex high, k is 8.38 and keys 11 of 15.

That term in mergers (UX6, MG1). On the Bullet's map [10], all seven readings read DEAD, both footings, both k. Local forms supply about the right mass, M_{300} 4.92, 2.44 against 3.5, 2.3 ($10^{14}\,M_\odot$), in the wrong place: $\Delta\kappa$ at the subcluster goes from −0.111 on the law to −1.158. Observed peaks sit on the galaxies. Only gas memory carried with the galaxies gets the sign right, and it fails on the main peak and on mass, and owes a carrier nobody has named. MG1, PROVISIONAL, from published numbers read through a web summariser: 8 of 9 subclusters that can decide have peaks on galaxies, 0 on gas, 1 between, in 6 systems. The sample was chosen for visible offsets, so that isn't a rate. Another reading of one criterion gives 7 of 12, and rows with published errors, not at the edge, number 4 in 3 systems. Abell 520 is contested. It isn't a map-level test.

A time lag in that term (TM2). Memory at past gas positions in a fixed frame reads DEAD at every lag on the Bullet, both footings. Vector memory carried along galaxy worldlines gets peaks and sign, misses the main mass at fitted k (Z +4.03, +3.89; alt +3.77, +3.63), and passes only post-result, at a fitted lag of 300–600 Myr and k of 7–10 against X-COP's 13.29, with 7 the edge of the scan. Stored per star, orbits erase it and relaxed clusters fail. Stored on the mean flow, discs spin up in a median 1.4–2.7 Gyr, an estimate on rotation-curve points. A third reading stays open and owes the k gap and an action. Its check K1 failed as frozen, so by its own rule these labels aren't a clean verdict.

k itself (KS1). $(M_{\rm tot} - M_{\rm law})/M_{\rm gas}$ is NOT ONE NUMBER, χ^2 473/14 (alt 528/14). It falls with radius, slope -0.52 ± 0.03, KiDS alone -0.45 ± 0.21. HSC reads +0.21 dex above KiDS, 3.8σ, 2.3σ with its own mass budget. The mass-dependence label comes from HSC alone, and we don't read it as one. A gas zero point of $+0.090 \pm 0.014$ on 5 X-COP clusters is unresolved. Total error is 25 % at $0.6\,R_{500}$ and 38 % at R_{500}.

Can't decide. Temperature or gas as carrier of the deficit: NOT SEPARABLE, both footings, F 0.52, 0.60 (alt 0.53, 0.55) against a line of 3 (TG1). Binary-pulsar accelerations [12] on AQUAL against the round rule: NOT SEPARATED, $2\Delta \ln L$ −0.98 (alt −1.24) on 23 pulsars, with mock power 0; about 1200 (alt 940) would be needed (PA1).

Published AeST (UX2). This is the published theory [4], not our version. It fails rows 3 and 6 at every published configuration, Υ 0.411/0.377 and IC 1613 E −0.618/−0.655. That failure is weak, since both come

from the MOND limit, and most other rows are CONDITIONAL or UNKNOWN. Its CMB fit holds at $g = 1$; at the physical g the benchmarks fail BAO and BBN. It reaches the CMB by carrying a fitted dust, ω_x 0.1206, linearly $w \approx 0$, $c_s \approx 0$. So at linear order it isn't a different answer from the retired sector. One check failed, 2.16 against 0.5, voiding the lane's "how dust-like" scan. What that dust does in nonlinear wells has no verdict, and that is the whole of row 4. A 1D reduction of the one-field action (AN1) withdrew its first headline after a centre-cell error. Rerun, 61 of 68 rows fold first, 5 never collapse, 2 oscillate and none settles, with no verdict because one frozen check missed. A 3D engine (AN2) has no verdict either: five frozen checks fail and a sixth misses. Diagnostics frozen before their runs read the grid-corrected shapes as folding 5–16 % before AN1's sphere, and that correction predicted fresh 160^3 rows within 0.7 % (SG, P160). These are reads, not labels.

Searching for an action (UX1, AS2, AS3). These results are empty for the listed candidates and the grammar, and neither is a no-go theorem. Of ten candidate classes, none structurally different passes; the one CONDITIONAL LEAD is v3s plus a separate dust, whose double count stands (capture 0.76). BIMOND [11] passes Q_2 with no screen, 0.029–0.107 of the line, and its tail still misses planets by 1.0e4 to 6.4e4 times their bounds. AS2 enumerated 2.17e11 raw combinations against analytic gates, no data: Tier A 0 of 717. 15 Tier B rows have no fail, and none is clean. The new corner it named, a khronon plus a scalar with $(D^2\chi)^2$, doesn't survive AS3: Q_2 and Ostrogradsky pass, but FRW growth fails, σ_8 ratio 0.0757 (alt 0.0817) against a line of 0.95, and so does the negative-charge instability.

Can one cold component do it (MC1, PC2)? On listed processes, MC1 ENDS AT REQUIREMENT 4. In budget, CMB and clusters plain dust is enough. Below $250\,\mathrm{km\,s^{-1}}$ it must be 66–82 % gone from discs with the measured response, and no computed process does that while it stays dust in the CMB. The required ratio to baryons is about zero in discs, 2–7 in clusters, 5.35 in the CMB. Density can't switch it: X-COP's need, $7.4 \times 10^{-4}\,M_\odot\,\mathrm{pc}^{-3}$, lies inside the disc range. A caution: a fixed-Υ pooled gate on record reads otherwise, the two gates disagree, and the verdict rests on our choice of gate (DW2 below traces that to the profile). A one-field phase change (PC2) is KILLED AT GATE 1 as a derivation, smallest miss -0.89 dex. With a counted mass of 3.74 eV it lands by fit, separates discs from clusters only marginally, $+0.09$ dex, and FRW thermalises. The only microphysics there that lands is particle-like, reported and not adopted.

That wall, and a max rule (DW2, MX1). These use a cold dark energy term as a test and reinstate nothing. How much must go depends on Υ: at 150–250 $\mathrm{km\,s^{-1}}$, 53–79 % at 0.50, 0–39 % at the lowest measured values, where the frozen label is NOT DECIDED. So MC1's wall is CONDITIONAL on a disputed stellar mass; no row reads WALL OPENS, and with cold in the law's source it holds throughout. The gate disagreement is the profile: fixed-Υ rms reads 0.1025 on a profile holding about 5 times less mass and 0.1298 on the native one, line 0.110, and no process makes the former. Of 15 processes, none passes every gate; gravity-only rows miss by more than a factor of 2, and the nearest are particle-like. Separately, $g = \max(\text{law}, g_b + g_c)$ removes nothing and read 13–14 of 15 keys after two post-result changes. An independent freeze, not blind, FAILS: keys 13 of 15, G-OUT-M3 +3.80 (alt +3.28), joint p 0.017 (alt 0.028), Bullet subcluster M_{300} 1.15 against 2.3. X-COP's pass, rms 0.0964, is marginal, and it turned on a no-expansion clamp unstated in DW2's solver, found through MX1's missed check K2. The vector rule has no parent in the class tested. Its aligned parent keeps momentum and breaks weak equivalence between cold and baryons: galaxies drift 2.9–4.3 R_{200} from their cold in groups within 3 Gyr, and its own $f = 0$ cell fails X-COP, 0.119. Keeping universal fall makes the force one-way instead. MX1's map at the switch isn't converged (K7), and it has no action or stated reason.

Searching for a formula (BF1, BF2, SR3, BK2, AF1). None found, and none of this proves absence. BF1 level L3: 2.102e9 expressions, 0 Stage 1 survivors, as in a scrambled control, and a planted form recovered. Its level L4, 7.5e11 expressions, also reads NO HIT: 0 Stage 1 survivors, as in its scrambled control, with no planted or MUTATE run at that level. Its near-miss family, law $\times(\mathrm{const} + \pi^2 g_{\mathrm{gas}}/a_0)$, fits X-COP out of sample and, on BF1's fixed-Υ statistic, breaks discs, rms 0.22–0.24 against a line of 0.175. AF1: NO FORMULA TRANSFERS. No clean variable carries the law's residual across samples, 0 CARRIES ACROSS on both footings, and SPARC has no survivor. Its frozen power plant failed and was replaced after the result.

BF2 took 3,952 hits from a snapshot of that running level and regraded them on FG1's free Υ, where 715 hold both rms and Υ (canonical), at twice or so the law's scatter in gas-rich slow galaxies. With no fitted float, two forms read NEAR MISS with 14 of 15 keys on both footings, as post-result picks, and a frozen-rule pick reads 15 of 15 and misses Υ at 0.47, 0.45. They earn no weight. The search is NOT ABOVE CONTROL: 19 real hits against 0, 15 and 266 for three scrambles on the same jobs, ratio to the mean 0.20. All three fail galaxy lensing, χ^2 845–885 on 56 against the law's 154 (canonical). All 24 present-gas Bullet cells read DEAD. No second implementation exists.

SR3, symbolic regression on the same free Υ with up to 2 fitted constants: NO CLEAN FORM, 0 of 141 real rows pass discs at the frozen line. A scrambled f reads 0.0554 against real 0.0567, and a pure cluster label passes every gate, so f acts as a sample label there. **Planted recovery failed** both frozen lines, so "nothing found" here isn't a proven absence, and any form on these fronts is a fit, not an identification. On the Bullet its graded forms have about the right mass and $\Delta\kappa$ wrong-signed at both subclusters, in all six rows.

BK2 enumerated dimensioned expressions, 2.7×10^{10} per footing at level L1: NO HIT on both footings, 0 through G1–G3 on the line or within errors. Its error reading of G3 is, by its own account, too strict, so the line is the fairer count. Closeness earns nothing: the scrambled control comes as close on both footings, 1.01 on canonical and 1.02 on alt against the real 1.08 and 1.07, where 1 is on the line, and canonical's control has no train pass among its 50 closest. The closest no-fit row reads 15 of 15 keys on the one cell run. Its content is that total mass follows the gas, and it fails SPARC, dwarfs, IC 1613, DiskMass and the Bullet's sign. Level L2, 7.4×10^{11} expressions per footing, also reads NO HIT: 0 through G1–G3 on the line or within errors, on both footings. Its closest rows read 1.01 and 1.01, and the scrambled control's 1.01 and 1.02, so the control comes about as close. Temperature and dispersion aren't in the grammar, and check K3 failed on the IC 1613 row.

The a_0 formula: no statistical support that it is special (BK1, CG1). The record called the formula certified by a gate. That pass was a unit error. Repaired, it reads 5.51 bits at 5 %, under the 10-bit line, FDR-DEAD, and the relation doesn't land in that window on either footing. On a real power test at SPARC's 16.6 % half-width, 92.7 % of scrambled scalars decompose as simply as the formula, 0.11 bits. Among rivals on SPARC its coefficient is best in 1 of 6 cells, by 0.24 bits. A port of the search (BK1) tried 5.238e9 trees, with hits 1.03 times a scrambled control and 0 passes. This isn't a refutation. The test lacks power at this precision: 10 bits would need a half-width of 0.0032 %, about 5000 times sharper, so a true formula couldn't earn evidence today either. What's left is the old nearness of a_0 to cH, which rivals share, and the law's performance on data, which these lanes don't grade.

Routes to $\kappa=\frac{1}{2}$ (KF1). Four proposed routes: NOTHING FORCED. Two give no number. One gives 1/2, 1.447 or 1.772 by the length chosen, and only the length with no field equation behind it gives 1/2. One trades κ for $\kappa=1/(2\xi)$. Forcing $\kappa=\frac{1}{2}$ is the same as forcing a length of exactly $c/\sqrt{G\rho}$. Five later routes, by hand algebra outside the frozen tests, also force nothing.

What the graded lines measure (DA1). VERDICT MOVES, PARTLY AUDITED, on lines only. Taken one at a time, other readings move cells of Table 3. With KiDS's bin $n(z)$ in $\Sigma_{\rm crit}$, C2's K reads 15 of 15. On a window of 0.5 ± 25 %, C1's S fails and the alt S of C3 and C4 passes; the 0.50 line's lower edge is the sources' centre, and no source read gives its 0.80. With SPARC's flow distances at native h, the S of C3 and C4 fails at 0.47, 0.48. Priors read as a mass scale put C1 (one background) and C3 inside on the Milky Way. X-COP's +0.111 mixes apertures: at one aperture it reads +0.092, 14 of 15 keys at f_* 0.034 and 15 at 0.0615, while the measured scale holds, at most +0.023, 12 of 15. Wall 1's f_Q and RM2's f aren't the same quantity; on RM2's own start the mass form reads 0.817 [0.454, 0.976], still above what RM2 needs. Each candidate still fails at least two lines on each reading, by inspection, since readings weren't run in combination except one pair. Several moves hurt the rule, and many inputs read CANNOT CHECK. These moves go to the owner.

A list of 96 rows (RQ3). A harness now grades any g on one list. Every candidate reads REJECTED. Known answers 110 of 110, after a first run at 103 of 110.

Table 5: RQ3 counts. They don't rank: rows are correlated in blocks, many are disputed, candidates compute different rows, and no cross-block covariance is on disk. Newton's 10 passes measure how little those rows ask.

candidate	footing	pass	fail	form-dep.	computed	keys
Newton	both	10	38	0	48	4 of 15
a_0 law	canonical	22	25	2	49	10
a_0 law	alt	20	27	2	49	10
UX3 term, k fit	canonical	29	15	0	44	14
UX3 term, k fit	alt	25	19	0	44	10
cold halo, known-answer test	both	5	6	0	11	not computable

For the law, 49 rows compute from g, 8 are READ, 1 is OPEN DATA and 38 are NOT COMPUTABLE, mostly for want of an action or a Boltzmann run.

Lanes since the first v1.1 draft (Table 6). Lanes frozen from 2026-11-02 to 2026-11-04 are listed with their own labels and the commits that carry their results. Where they change a requirement row, the sections above

and Table 7 carry the change. The rest follows the table.

Rules on the law itself (CX1, NP1, NP2, NV1, NV2). These ask whether one rule built from local quantities, g_b/a_0, a length and the gas share, could fit every row. In that space no two systems need different boosts beyond their errors, except lensing against SPARC, and that rests on the innermost lensing radius (CX1). X-COP has no near neighbour, so a rule can lift it without moving any other row. A smooth rule fitted on train alone holds the unseen X-COP clusters, +0.003 dex, and the held-out halves of SPARC and LITTLE THINGS on both footings (NP1). The 96-row list rejects it in every cell. Kept admissible, with g rising with g_b, it still fails IC 1613 and the Υ rows inside its training range (NP2). No non-local variable on record resolves those rows. Basin mass moves the dwarfs and lensing partway and breaks SPARC (NV1), and the external field computed for X-COP gives an inadmissible fit (NV2). A rule that passes here describes data. It isn't a theory.

Added components (DF2, DF3, RS1, WF1). These ask what a component beside the law would need, and they adopt nothing. A barotropic fluid keyed to density passes the early universe in 5 of 10 cells (DF2). None of those passes discs, clusters or the Bullet: $\Gamma = 5/3$ collapses into a core holding the whole fair share, and $\Gamma \leq 1$ has no equilibrium in clusters (DF3). The extra source the law would need doesn't collapse on any one variable, and 3.5e9 forms come no closer than about twice the lines, with X-COP and the ultra-faints as the wall (RS1). A wave-pressure field passes the early universe from 10^{-25} eV up, and at that mass it fails the cluster keys, 1 of 15, and the groups, Z −8.31 (WF1).

Estimators, not gravity tests (LAM1–LAM9). These build spin and stellar-population labels for a later lensing split. Where they test the law's residual against orbit shape, they read NO VERDICT or h4 NOT MEASURABLE (LAM6–LAM8), and LAM9 is still running.

8 What a successor must do

Table 7 lists twelve requirements.

The collisions. Rows 4 and 11: dust in the CMB falls into galaxies. Rows 1–3 and 10: the amount needed runs from about 0.8 of native on SPARC to 1.22–1.80 on clusters. Rows 7 and 8: systems of the same mass want different things. Selectors closed on inputs tested: potential, σ, acceleration, environment, formation epoch, gas state, bars. None of this is a theorem.

One family left outside, then tried once. A 5–11 eV fermion would address rows 4 and 11 directly. Its forest kill isn't on the current footing: a 1-D proxy against a ΛCDM yardstick falls 1000–5000 times short, undecided on raw forest data. Other liabilities stand, and it wouldn't fix rows 6 or 7. It's outside the no-particle rule. The owner lifted that rule for one experiment lane only (HF1), which adopts nothing; the rule stands everywhere else. A thermal fermion beside the law fails the early universe at 5 of 6 masses, best plik 1432.4 on 605 points at 11.8 eV. The sixth, 3.74 eV, didn't converge, so the lane doesn't call the family dead outright. As numbers only, it also fails X-COP, +0.159, the cluster keys, 3 of 15, and the groups, Z −5.90. A slower species doesn't help (VW1). The early universe allows at most 2–4 $\mathrm{km\,s^{-1}}$ today, SPARC on canonical needs between 6.72 and 14.9, and that gap rests on the Υ floor of 0.50. Opening the floor wouldn't save it: at 50 eV clusters and groups still fail, keys 3 of 15.

9 Dated tests, and their real power

- **ALMA U4_28473, public 2026-11-06 02:42:31 UTC** (PAPER66 v1.4, DOI 10.5281/zenodo.23958256). It tests this galaxy's mass budget, not $a_0(z)$, and one disc excludes no model.
- **GS4_24110,** [CI](2–1), 2026-11-28 by earlier metadata, to be rechecked. Every $n = 1$ against $n = 2$ pair is at or below 0.51σ, so it can't separate them.
- **Gaia DR4, about December 2026** (pre-registered 2026-07-16, with later amendments). Wide binaries: Newton 1.000, an ungated MI band 1.05–1.10, AQUAL point 1.137, MOND's benchmark 1.33; the DR3 dry run's 1.205 ± 0.035 is evidence for nothing. An amendment registers a gated branch for which 1.000 inside 30 kAU is also predicted, so that branch needs its own test beyond 50 kAU, about 2000 clean pairs per bin. The solar-acceleration dipole's kill bands are void under the amended theory, so that front returns no verdict. *Reading, not a frozen line:* these bands were written for the action-carrying framework. The bare law here contains no gate, so a Newtonian result at 2–30 kAU would count against it. A sealed prediction (CW1) puts the law in QUMOND form, unscreened, at γ_∞ 1.149 (alt 1.208) through the frozen pipeline, and the failed

Table 6: Lanes frozen from 2026-11-02 to 2026-11-04, by each lane's own label, with the commit that carries its result. None was blind, all were run by AI subagents, and none reinstates the cold sector. HF1 and VW1 ran under the owner's lift of the no-particle rule for that experiment only, and they adopt nothing.

lane	idea	label	commit
HF1	light thermal fermion beside the law	FAILS GATE 1 AT 5 OF 6 m, 1 UNDETERMINED	93ab39c78a
VW1	velocity window for that fermion	WINDOW EMPTY on canonical, on the Υ floor	086516fd25
DF2	barotropic dark fluid, early universe	DENSITY KEY PASSES GATE 1, one gate only	2d73d459fb
DF3, two lanes	those fluids in discs, clusters and the Bullet	NEITHER; NO GATE	1139b88abb, 9f549e8963
RS1	the source the law would need	NO SOURCE FORM	36742f590f
WF1	wave-pressure scalar field	SURVIVES TO gate 1; gates 2–3 fail	d390fc1580
CX1	collisions in leaf space	NO-GO IN LEAF SPACE, only through lensing against SPARC	1cba79837d
NP1	smooth leaf rule fitted on train	gates 1–5 pass; REJECTED by the 96-row list	811d215010
NP2	that rule kept admissible	ADMISSIBLE RULE FAILS ON A10, A5–A7 and more	b1bb9b29d9
NV1, NV2	one non-local variable; external field from 2MPZ	NO SINGLE NON-LOCAL VARIABLE ON RECORD RESOLVES	e5371291f9, 472328c26c
BF1 L4	closed forms with no fitted float	NO HIT	2654bb5551
BK2 L1	canonical's scrambled control	as close as the real data	aeec6c0283
BK2 L2	dimensioned expressions, level L2	NO HIT at L2 on both footings	eeb2f01037
TX1	temperature as a leaf variable	SELF-FIT ONLY on both footings	3ae8ffb2d1
MB1	member-bound law halos as the cluster variable	no hit, REJECTED in every arm on both footings	8f23cc6c57
BF3	every gate first, one joint form	JOINT FORM, NOT ABOVE CONTROL; fails the holdouts	648ec0ebc6
DB1	Milgrom's cluster dark baryons	NO CBDM READING PASSES	40e2665f48
FN1b	one universal smoothing length, five static gates	NO UNIVERSAL LENGTH PASSES	22f8e50070
FN3	MUSE-DARK cubes refitted with the law imposed	NOT DECIDABLE WITH THE EXISTING CUBES	5c1880e6c6
FN3b	the same with HI fixed from local HI structure	NOT DECIDABLE	cec5a0337d
FN4b	Local Group external field from UNGC	FIELD MOVES NOTHING	79658c0ec8
FN4c	Brouwer+21's isolated lenses rebuilt	REBUILD MATCHES; GAP STANDS	ae9bd592c8
FN4a	IC 1613's HI cube fitted with the law	NOT EVALUABLE: the cube fails its stage-0 checks	db14eb9fac
FN4f	Milky Way thick-disc share from APOGEE DR17	NOT MEASURABLE without the selection function	bbe016eff5
LE1	lensing estimators	ESTIMATORS DISAGREE, NO DEFECT	17574fce14
CW6	lensing with measured kSZ gas	GAP UNCHANGED	b01c37cdc7
WG1	the potential CMB lensing maps, against a_0	inside the law's range, 0.003–0.22 a_0	93e5f4e99c
WG2	the law alone on the web from a baryons-only start	FAILS under every web rule tried; NOT EVALUABLE as frozen	6856666719
WG3	the record's bound-only switch on the web	FAILS LOW; it never turns on	d4d81be402
WG4	a_0 tracking evolving dark energy on the web	FAILS HIGH as frozen; same shape at about half	5cd2df9f68
WG6	a screening term on the law's extra mass, by hand	WITHIN x2 at 100 Mpc as frozen, but the shape fails	628285f8c2
KT14–KT32	phantom mass smoothed over a length	no form passes every row; tidal line FAILS MW	a3441c7725 to ffa3e14d39
CZ1–CZ4	Newtonian external field in ν; regrades	NUMBERS ONLY (CZ1); survivals on alt in 1D	8e701cc3e6 to 351db5b6de
CZ5	the local rule on the 3D field	LOC FAILS ON BOTH FOOTINGS	528285bafe
CW1	wide-binary seal for Gaia DR4	SEALED	e7672a30e1
CW3	external-field sweep of the record	ONE LABEL FLIPS (KT30); none in this paper's rows	5540d45cb5
AX1	every action class against this law	NO UNSCREENED ACTION GIVES THE LAW AND PASSES THE SOLAR SYSTEM	9c11102bf3
C6P	v3s's failed check at high precision	PASS on both footings	8c5703da06
KT33–KT42	smoothed and curvature responses, full adjoint	NO VIABLE RESPONSE; the four-cut and Q_2 pass fails force sign, dwarf virials and planets	c1ac71c574
VH1	v3s through KT40/KT42's three health gates	PASS on both footings; adopts nothing	0c677158c7
V3W	v3s's weak spots: PPN, lensing, GW speed, health	W1, W3, W4a/b PASS; W2, W4c UNDECIDED	17d7b7ecdc
V3X	closing V3W's two UNDECIDED	W2 UNDECIDED (reading PASS); W4c (a),	ee4210e959

Table 7: Requirements from combining the failures. Older rows are on older chains, and FG1 supersedes any it covers. Coverage: an inventory that opened about 90 of about 330 lanes.

#	must	law	native halos
1	keep galaxies tight	**0.0541**	0.0682 / 0.0694
2	stay flat with surface brightness	−0.016	−0.173
3	sit on Υ at every speed	0.52, alt 0.44	0.75, running 1.20 to 0.57
4	show little net contraction in spirals	not needed	needs $f \leq 0.2$, measured 0.872; beside the law, 66–82 % gone (MC1), conditional on Υ (DW2)
5	fit dwarfs	−0.025; ultra-faints still +5.08 (UX4); filters don't lift them (CZ5); the Local Group field moves none (FN4b)	+0.138; 0 of 5 classes
6	fit IC 1613	−0.593	−0.474
7	fit the Milky Way	AQUAL, QUMOND INFEASIBLE; round rule only; no filter form survives (CZ5)	asks 4e10 $M_\odot$ inside R_0
8	fit isolated lenses	shape preferred; 0.09 dex short inside 0.3 Mpc (LE1, CW6); stands on the rebuilt Brouwer sample (FN4c)	+366.0
9	add mass in cluster cores	A 1.91	G-IN −0.49, G-MID +2.17
10	match outer cluster shear	G-OUT-M3 +4.76; slope 0.47 against 0.90	G-OUT-M3 +4.41
11	act as dust in the CMB	no cold term	$z_{\rm on} \in [3 \times 10^4, 10^6]$; ω_x 0.1197 (UX2)
12	have an action	met in a scope only; 0 of 717 in AS2's Tier A; unscreened classes fail Q_2 (AX1); only v3s's screen passes the Solar System (C6P); no one-length smoothing passes (FN1b), nor any curvature response (KT33–KT42); v3s passes their health gates (VH1); **v3s adopted, scoped** to galaxies, the Solar System, lensing and GWs (owner, 2026-11-04, Table 2); clusters and the CMB open	$P(X)$ dust limit only

local filter rule at 0.986 (alt 0.981), which that pipeline can't tell from Newton.

- **SP1, SP2 sealed lensing predictions.** Scored as frozen, including the retired route's rows.
- **Named, not scheduled:** one clean deep-MOND a_0 point at $z \approx 2.5$ to about 0.10 dex. That's the measurement that would actually test the dark energy tie.
- **Run as FN3, 2026-11-04:** a per-galaxy MUSE-DARK refit in 3D with this law imposed and a stated baryon-mass budget. It can't decide: with HI unmeasured, a planted rise of the survey's size grades BAND CONSISTENT on both footings (slope recovered +0.32 of 0.60 per unit z). Fixing the HI term from local HI disc structure (FN3b) still leaves a bias of up to -0.16 dex and recovers the rise at +0.47 of 0.60, so the test waits on deeper kinematics as well as measured gas.

10 Method

Each lane commits frozen criteria alone before any script or number, runs a MUTATE twin that must fail, reruns byte-identically and publishes failures under frozen labels. Most lanes were not blind. Grading was done by the same programme that proposed the models; the clean-room regrade tested code, not independence of definitions. No outside group has reproduced these results. Lanes in Section 7 were run by AI subagents after the retirement, and their failed checks are listed in each README.

Not claimed

A complete, closed or unified theory. That the data favour this framework. κ, T or any abundance as derived. That the tie of a_0 to dark energy originated here. That the requirements list is a no-go theorem. That v3s's adoption completes the theory: it is adopted as the law's action only for isolated galaxies, the Solar System, galaxy lensing and gravitational waves, it supplies no cluster or CMB mass, and clusters, growth and the CMB stay open.

Data and code

All lanes named here are in `rebuild_2026_10_22/` and `campaign_fresh_gravity/`, except CZ1–CZ5 in `claude_goal_2026_11_03/` CW1, CW3 and CW6 in `cw_lanes_2026_11_04/`, and FN1b, FN3–FN6 and the owner's action decisions in `finish_2026_11_04/`. The κ closure is `campaign_fresh_gravity/CFG894_kappa_campaign_ledger/KAPPA_CLOSURE_2026-11-03.md`. Scorecard: `FG1_final_grade`. Law's scorecard: `LW1_law_scorecard`. Ledger, still a working draft: `CLOSING_LEDGER_2026-11-01_DRAFT.md`. Lanes new in v1.1 are in `rebuild_2026_10_22/`, under the names in Section 7: `AR1_action_response`, `UX3_cluster_gas_term`, `CU1_gas_term_replication`, `CG1_a0_certificate`, `RQ3_acceptance_v2`, `BK2_dimensional_law_search`, `DA1_definition_audit` and others. A companion ledger for the rules on the law, still a working draft: `LAW_RULE_LEDGER_2026-11-04_DRAFT.md`. BK2's README is the record of both levels. Audit: `PAPER67_audit.py`.

References

[1] M. Milgrom, ApJ 270, 365 (1983).

[2] M. Milgrom, Phys. Lett. A 253, 273 (1999).

[3] C. Limbach, D. Psaltis, F. Özel, arXiv:0809.2790 (2008).

[4] C. Skordis, T. Złośnik, arXiv:2007.00082 (2020).

[5] L. Blanchet, C. Skordis, arXiv:2404.06584 (2024).

[6] A. A. Dutton et al., arXiv:0801.1505 (2008).

[7] S. Aniyan et al., MNRAS 456, 1484 (2016), arXiv:1511.06047.

[8] S. Aniyan et al., MNRAS 476, 1909 (2018), arXiv:1802.00465.

[9] S. Aniyan et al., arXiv:2010.03991 (2020).

[10] Clowe et al., arXiv:astro-ph/0608407 (2006).

[11] M. Milgrom, arXiv:0912.0790 (2009).

[12] Donlon et al., arXiv:2501.03409 (2025).

[13] B. I. Ciocan et al., A&A 709, L16 (2026), arXiv:2604.22613.

[14] G. T. Alcock, MNRAS 551, L1 (2026), arXiv:2609.38796.

[15] Jeanneau et al., A&A 709, A120 (2026), arXiv:2603.28856.